Questions

Q1.

Everyone is exposed to background radiation. Some of this radiation comes from natural sources.
* There are many radioactive isotopes in nuclear waste.

Technetium-99 is just one of these isotopes.

People are worried about how we should deal with nuclear waste.

Explain why it is difficult to deal with nuclear waste safely.

(6)

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Q2.

This question is about radioactivity.

* Exposing people to radioactive sources can be dangerous.

Describe the dangers of exposure to radioactive sources and what can be done to protect hospital staff when they are working with radioactive sources.

(6)

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(Total for question = 6 marks)

Q3.

The fuel in a nuclear power station is an isotope of uranium.
Control rods are used in the nuclear reactor.

Explain how these rods stop the nuclear reaction from getting out of control.

(2)

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Q4.

Figure 6 shows a diagram of a helium atom.

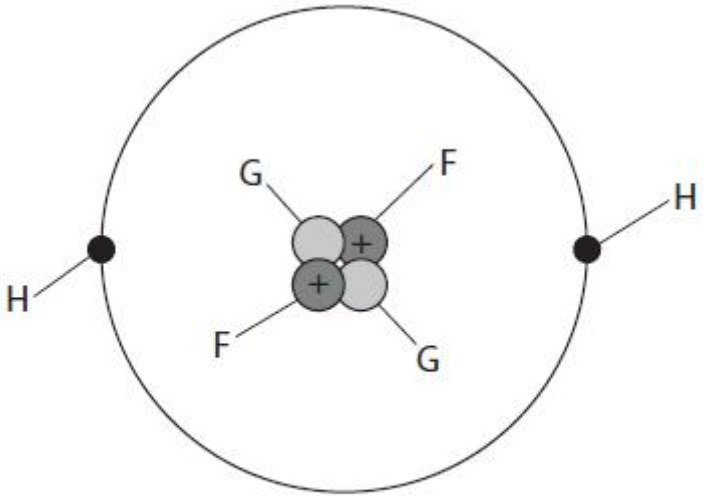

Figure 6

Complete the table for particles G and H, as shown in Figure 6.

(4)

particle	name of particle	charge
F	proton	positive
G		
H		

(Total for question = 4 marks)

Q5.

Carbon-13 and carbon-14 are isotopes of carbon.

Nuclei of carbon-13 and carbon-14 can be represented by these symbols

$$^{13}_{6}C \quad \text{and} \quad ^{14}_{6}C$$

Complete the table for an atom of carbon-13 and an atom of carbon-14.

(2)

	number of neutrons in the nucleus	number of electrons in orbit around the nucleus
carbon-13		
carbon-14		

(Total for question = 2 marks)

Q6.

The half-life of strontium-90 is 29 years.

The table in Figure 9 gives some information about how the mass of a sample of strontium-90 changes with time.

mass of strontium-90 in g	time in years
1600	0
.....	29
400	.....

Figure 9

Complete the table in Figure 9.

(Total for question = 2 marks)

Q7.

Use words from the box to complete the sentences about nuclear fission of uranium-235 (U-235).

chain	chemical	fuse
neutrons	protons	split

(3)

A neutron hits a nucleus of U-235 and causes the nucleus to ..... .

Each fission releases energy, two daughter nuclei and some ..... .

In a nuclear reactor, one fission can set off a controlled ..... reaction.

(Total for question = 3 marks)

Q8.

This question is about radioactivity.

Fluorine-19 is a stable isotope of the element fluorine.

The element fluorine also has several radioactive isotopes.

Describe **one** similarity and **one** difference between the numbers of particles in one nucleus of fluorine-19 and one nucleus of a radioactive isotope of fluorine.

(2)

similarity

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.....

difference

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(Total for question = 2 marks)

Q9.

Radioactive tracers can be used when scanning a person's kidneys.
A radioactive isotope is injected into a person's blood stream.
The isotope emits radiation.
As the blood flows through the kidneys, this radiation is detected outside the body by a scanner.

(i) What type of radiation travels from the kidney to the scanner?

(1)

- ☐ **A** alpha
- ☐ **B** beta plus
- ☐ **C** beta minus
- ☐ **D** gamma

(ii) During the scan, a technician needs to take readings for about 30 minutes.
The half-life of the isotope used is about 6 hours.
1. State why an isotope with a half-life of about 6 minutes is **not** suitable.

(1)

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2. State why an isotope with a half-life of about 6 days is **not** suitable.

(1)

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(iii) State **two** ways of reducing the radiation risks to the technician.

(2)

1 .....

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2 .....

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(Total for question = 5 marks)

Q10.

In 2023, Finland produced 4.4 GW of power from nuclear power plants.

The UK produced 8.9 GW of power from nuclear power plants.

Calculate the ratio of nuclear power produced in Finland compared with the UK as

$$\frac{\text{power production in Finland}}{\text{power production in the UK}}$$

Give your answer to 2 significant figures (2 s.f.).

(2)

ratio = ...

(Total for question = 2 marks)

Mark Scheme

Q1.

		Indicative Content	Mark
		an explanation which may include some of the following points: properties of nuclear waste radioactivity is dangerous some isotopes in nuclear waste have long half-lives/radioactive for thousands of years products of fission are warm identified radiation from nuclear waste e.g alpha, beta, gamma problems caused by nuclear waste leakage of radioactivity contamination of ground/sea water/lakes /rivers contamination of crops/fish/animals/drinking water harm to humans/cancer/radiation poisoning/ damage to cells/mutation of cells or DNA difficulty in transporting safely/ stolen by terrorists fears of local people solutions for dealing with nuclear waste safely long term storage, underground /under the sea radiation shielding, lead/steel/concrete/ containers, sealed in glass. human safety, radiation suits, using tongs/lead jackets safe location, away from people/remote	**(6)**

		areas/sea cooling, ponds information to persuade local people of safety	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited explanation mentioning at least one point, but without linking, e.g. radioactivity is dangerous ; nuclear waste should be stored underground ; terrorists might steal nuclear waste; • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple explanation mentioning two points with an appropriate linkage e.g. nuclear waste is dangerous and it must be stored underground ; the isotopes in nuclear waste have long half-lives so they must be stored for a long time; • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed explanation mentioning a range of points with appropriate linkages e.g. gamma rays from nuclear waste causes damage to cells so it must be stored away from where people live ; the isotopes in nuclear waste have long half-lives so they must be stored underground or in remote areas; • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q2.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **Dangers of exposing people to radioactive sources/radiation.** • it is ionising • may cause cancer • may destroy /kill cells • can mutate DNA • can burn the skin **Protection of hospital staff using radioactive sources/radiation.** • use tongs to carry radioactive sources • use lead containers to store sources • stay at a distance from radioactive sources • use sources for as short a time as possible • wear (lead lined) protective clothing (PPE) • give treatments from behind a shield /wall • wear a radiation badge (dosimeter)	**(6)** **AO1**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3–4	Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5–6	Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

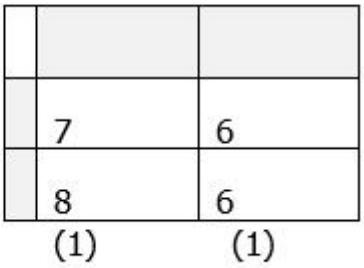
Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance At least one isolated fact about the dangers of radiation and/or protection from radiation	Possible candidate responses it's ionising causes cancer burns you kills cells mutates DNA wear a radiation badge use tongs work from behind a shield use protective clothing
Level 2	3–4	Additional guidance simple explanation of the dangers of radiation and a fact about protection or reverse OR detailed explanation of the dangers of radiation or protection from radiation	Possible candidate responses radiation is ionising and can kill cells so wear a radiation badge **or** use tongs and stay at a distance from radiation source as it can cause cancer **or** use tongs to stay at a distance from radiation sources and wear a radiation badge
Level 3	5–6	Additional guidance detailed explanation of the dangers of radiation and protection from radiation	Possible candidate responses radiation is ionising and can kill cells **and** use tongs and stay at a distance from the radiation source

Q3.

	Answer	Acceptable answers	Mark
	An explanation linking two of the following points • absorb (1) • neutrons (1) • (influences) chain reaction / rate of reaction (1)	Accept reverse arguments collects/removes/takes away slows down/changes	**(2)**

Q4.

Q5.

Question Number:	Answer	Additional guidance	Mark
	 7 6 8 6 (1) (1)	one mark for each column must have both numbers in a column correct to get the mark	**(2)**

Q6.

Question number	Answer	Additional guidance	Mark
	<table><tr><th>mass in g</th><th>time in days</th></tr><tr><td>1600</td><td>0</td></tr><tr><td>800 (1)</td><td>29</td></tr><tr><td>400</td><td>58 (1)</td></tr></table>	numbers in correct boxes	(2)

Q7.

Question number	Answer	Additional guidance	Mark
	split (1) neutrons (1) chain (1)	in this order	(3)

Q8.

Question number	Answer	Additional guidance	Mark
	same number of protons (1) different number of neutrons (1)	same atomic number different mass number	(2) AO2

Q9.

Question Number	Answer	Mark
(i)	**D gamma** ***A*** *is not correct because alpha cannot pass through and out of the body* ***B*** *is not correct because beta plus cannot pass through and out of the body* ***C*** *is not correct because beta minus cannot pass through and out of the body*	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii)	decays too quickly to give a reading (1)	accept (half-life) not long enough for reading to be taken Ignore disappear after 12 min	**(1)** **AO1**

	Answer	Additional guidance	Mark
(ii)	stays in the body too long (1)	accept could harm/damage other organs patients stay radioactive for too long so the patient does not get too high a dose of radiation	**(1)** **AO1**

	Answer	Additional guidance	Mark
(iii)	**Two** from: shielding (1) time limiting exposure (1) distance limiting exposure (1) wear PPE / protective clothing (1)	 accept stand behind barriers / store (source) in lead box radiation monitoring badges leave the room/ go outside/stay away from the patient / use tongs lead aprons / gloves ignore goggles / masks	**(2)** **AO1**

Q10.